WORKSHEET — UNIT V

Machine Learning for Data Science

Course: Data Science with Python **Unit:** V (12 Hrs)

Name: _____ **Roll No:** _____ **Date:** _____

Topics covered: Types of ML • Train-test split • Regression & classification • Clustering • Evaluation • scikit-learn

Section 1 — Key Concepts (Quick Revision)

Term	Simple Meaning	Example
Supervised learning	Learn from data that has answers (labels).	`spam / not spam emails`
Unsupervised learning	Find groups in data that has no labels.	`grouping customers`
Regression	Predicts a number.	`house price`
Classification	Predicts a category.	`pass / fail, yes / no`
Train-test split	Keep some data hidden to test the model.	`train_test_split(X, y)`
Feature scaling	Bring all features to a similar range.	`StandardScaler()`
Overfitting	Model is good on training data but poor on new data.	`it "memorizes" the data`
Cross-validation	Test the model on several splits for a fairer score.	`cross_val_score(...)`
Confusion matrix	Table of correct and wrong predictions.	`confusion_matrix(y_test, y_pred)`
Accuracy, precision, recall	Accuracy: % of all predictions correct. Precision: of predicted positives, how many are right. Recall: of real positives, how many were found.	`accuracy_score(y_test, y_pred)`
k-Means	Groups data into k clusters.	`KMeans(n_clusters=3)`
PCA	Reduces the number of features.	`PCA(n_components=2)`
scikit-learn	Python library for building ML models.	`from sklearn... import ...`

Section 2 — Multiple-Choice Questions

Choose the correct option and circle the letter.

1. **Which type of machine learning uses labeled data?**
   - **A.** Unsupervised learning **B.** Reinforcement learning
   - **C.** Supervised learning **D.** None of these
2. **Which algorithm predicts a continuous number (like a house price)?**
   - **A.** Linear Regression **B.** Logistic Regression
   - **C.** k-Means **D.** Decision Tree Classifier
3. **Why do we use a train-test split?**
   - **A.** To copy the dataset **B.** To test the model on unseen data
   - **C.** To remove outliers **D.** To scale the features

4. **Which technique reduces the number of features in a dataset?**
   A. k-Means **B.** Decision Tree
   C. Cross-validation **D.** PCA

5. **High recall means that:**
   A. Most predictions are negative **B.** Most actual positives were found
   C. The model always says positive **D.** The model is overfitting

6. **Which algorithm groups data into clusters without using labels?**
   A. Logistic Regression **B.** Decision Tree
   C. k-Means **D.** kNN

7. **What does overfitting mean?**
   A. Good on training data, poor on new data **B.** Poor on both training and new data
   C. The model has too few features **D.** The model cannot be trained

8. **Which metric is the percentage of all predictions that are correct?**
   A. Precision **B.** Recall
   C. F1-score **D.** Accuracy

9. **Which Python library is commonly used to build ML models?**
   A. BeautifulSoup **B.** scikit-learn
   C. Tkinter **D.** SQLAlchemy

10. **Predicting whether an email is spam or not spam is an example of:**
    A. Clustering **B.** Regression
    C. Classification **D.** PCA

Section 3 — Fill in the Blanks

Word bank: Supervised k-Means Overfitting confusion scikit-learn

1. _____ learning uses labeled data to train a model.
2. _____ is an unsupervised algorithm that groups data into k clusters.
3. _____ happens when a model is good on training data but poor on new data.
4. The _____ matrix shows correct and wrong predictions of a classifier.
5. _____ is a Python library used to build and test ML models.

Section 4 — True or False

1. Linear regression is used to predict a continuous number. **(True / False)**
2. k-Means clustering needs labeled data. **(True / False)**
3. Overfitting means the model works well on new data. **(True / False)**
4. Cross-validation gives a more reliable estimate of model performance. **(True / False)**
5. PCA is used to reduce the number of features. **(True / False)**

Section 5 — Match the Following

Write the correct letter from Column B next to each item in Column A.

No.	Column A	Ans.	Column B
1	Linear Regression		a. Table of correct and wrong predictions
2	Logistic Regression		b. Groups data without labels
3	k-Means		c. Predicts a number
4	PCA		d. Predicts a category (such as 0 or 1)
5	Decision Tree		e. Reduces the number of features
6	Confusion Matrix		f. Tree of yes/no rules

Section 6 — Short-Answer Questions

Answer each question in 1–2 sentences.

1. **Write one difference between supervised and unsupervised learning.**

2. **Why do we scale features before training a model?**

3. **Write the meaning of precision and recall in one line each.**

4. **Write one difference between k-Means and kNN.**

5. **What is overfitting? Give a simple example.**

Section 7 — Practical Programs

Write and run each program in Python (scikit-learn). Fill the blanks (___) using the hint.

Practical 1: Predict house prices (Linear Regression)

1. Make a small dataset with house size and price.
2. Split it into training and test sets.
3. Train a LinearRegression model and predict the test prices.

Hint:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

df = pd.DataFrame({"size":  [500, 750, 1000, 1250, 1500, 1750, 2000, 2250],
                   "price": [25, 37, 50, 62, 75, 88, 100, 112]})   # lakhs

X = df[["size"]]
y = df["price"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=1)

model = LinearRegression()
model.___(X_train, y_train)       # train the model
print(model.___(X_test))            # predict prices
print(y_test.values)               # actual prices
```

Expected output: *Predicted prices (about 112.7 and 49.8) close to the actual prices 112 and 50.*

Practical 2: Group customers (k-Means)

1. Make a dataset with customer income and spending score.
2. Apply KMeans with k = 3.
3. Print the cluster label of each customer.

Hint:

```
import pandas as pd
from sklearn.cluster import KMeans

df = pd.DataFrame({"income":   [15, 16, 60, 62, 120, 125, 18, 64, 122],
                   "spending": [80, 75, 50, 45, 20, 15, 78, 48, 18]})

kmeans = KMeans(n_clusters=___, n_init=10, random_state=0)
df["cluster"] = kmeans.___(df[["income", "spending"]])   # fit and get labels
print(df)
```

Expected output: *Each customer gets a label 0, 1 or 2. Similar customers share a label (the label numbers may differ on your computer).*

Practical 3: Check a classifier (confusion matrix, accuracy, precision, recall)

1. Train Logistic Regression on a small labeled dataset (hours studied → pass or fail).
2. Predict labels for the test set.
3. Print the confusion matrix, accuracy, precision and recall.

Hint:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score
from sklearn.metrics import precision_score, recall_score

df = pd.DataFrame({"hours":  [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],
                   "passed": [0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 1, 1]})

X = df[["hours"]]
y = df["passed"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.33, random_state=1, stratify=y)

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print(confusion_matrix(y_test, y_pred))
print("Accuracy :", accuracy_score(y_test, ___))
print("Precision:", precision_score(y_test, y_pred))
print("Recall   :", ___(y_test, y_pred))
```

Expected output: *A 2×2 confusion matrix, then Accuracy 0.75, Precision 0.67, Recall 1.0.*

Instructor Answer Key

Section 2 — MCQ

Q	Answer	Why
1	C	Supervised learning trains on data that has labels.
2	A	Linear regression predicts a number. Logistic regression predicts a category.
3	B	The test set shows how the model does on unseen data.
4	D	PCA reduces the number of features and keeps most of the information.
5	B	Recall = real positives found ÷ all real positives.
6	C	k-Means is an unsupervised clustering algorithm.
7	A	Overfitting = good on training data, poor on new data.
8	D	Accuracy = correct predictions ÷ all predictions.
9	B	scikit-learn has many ML algorithms and tools.
10	C	Spam / not spam is a category, so it is classification.

Section 3 — Fill in the Blanks

1. Supervised 2. k-Means 3. Overfitting 4. confusion 5. scikit-learn

Section 4 — True or False

1. True 2. False 3. False 4. True 5. True

Section 5 — Match the Following

1 → c 2 → d 3 → b 4 → e 5 → f 6 → a

Section 6 — Short Answers (Key Points)

- Supervised learning uses labeled data. Unsupervised learning finds groups or patterns in data with no labels.
- Features can have very different ranges (age vs salary). Scaling puts them on a similar range so one feature does not dominate.
- Precision: of all predicted positives, how many were correct. Recall: of all real positives, how many were found.
- k-Means is unsupervised and groups data into clusters. kNN is supervised and uses labeled neighbours to predict.
- The model memorizes the training data but does badly on new data. Example: 100% on training data, 60% on test data.

Section 7 — Practical Programs (Sample Solutions)

Practical 1

```
model.fit(X_train, y_train)
print(model.predict(X_test))    # [112.71 49.83]
```

Practical 2

```
kmeans = KMeans(n_clusters=3, n_init=10, random_state=0)
df["cluster"] = kmeans.fit_predict(df[["income", "spending"]])
```

Practical 3

```
print("Accuracy :", accuracy_score(y_test, y_pred))    # 0.75
print("Recall   :", recall_score(y_test, y_pred))       # 1.0
```